Heart Disease Prediction

Classifying heart-disease risk from routine clinical measurements

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0.85	0.91	0.52
Test accuracy	ROC-AUC	Baseline accuracy

1. Background and question

Can routine clinical measurements predict whether a patient has heart disease, and which factors matter most?

Heart disease is one of the leading causes of death worldwide, and much of its burden comes from cases caught late. Most risk factors (blood pressure, cholesterol, exercise test results) are already measured in routine check-ups. The opportunity is to combine them into a single, explainable risk score that helps clinicians decide who needs further testing first.

2. Data

- UCI Cleveland Heart Disease dataset: 303 patients, 13 clinical features, binary target (165 with disease, 138 without)
- Features: age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression, ST slope, number of major vessels, thal
- No missing values; one duplicate row; mean patient age 54.4

Data dictionary

Feature	Description	Type
`age`	Patient age in years	Numerical
`sex`	Sex (1 = male, 0 = female)	Categorical
`cp`	Chest pain type (0-3)	Categorical
`trestbps`	Resting blood pressure (mm Hg)	Numerical
`chol`	Serum cholesterol (mg/dl)	Numerical
`fbs`	Fasting blood sugar > 120 mg/dl	Categorical
`restecg`	Resting ECG result (0-2)	Categorical
`thalach`	Maximum heart rate in stress test	Numerical
`exang`	Exercise-induced angina (1 = yes)	Categorical
`oldpeak`	ST depression induced by exercise	Numerical
`slope`	Slope of peak exercise ST segment (0-2)	Categorical

Feature	Description	Type
`ca`	Major vessels coloured by fluoroscopy (0-4)	Numerical
`thal`	Thalassemia test result (coded 0-3)	Categorical
`target`	Heart disease present (1 = yes)	Target

3. Method

- Exploratory analysis: distributions, disease rate by chest-pain type, correlation matrix
- 70/30 train-test split plus 10-fold stratified cross-validation
- Models: majority-class baseline, decision tree, logistic regression (raw features), random forest, logistic regression with scaling and one-hot encoding
- Metrics: accuracy, ROC-AUC, precision, recall, F1; ROC curves; model coefficients for interpretation
- Generative AI extension (RAG): a 23-passage knowledge base is embedded with the all-MiniLM-L6-v2 sentence-transformer; each patient's top-3 factors drive retrieval; facts come from a template, definitions are quoted from the retrieved passages, and a local LLM (Qwen2.5-1.5B-Instruct via Hugging Face) answers follow-up questions in a Gradio chat app; five designs were evaluated

4. Project walkthrough

The problem. I wanted to know whether 13 measurements that hospitals already collect could identify patients with heart disease, and which of those measurements matter most.

The data. The UCI Cleveland dataset has 303 patients: 165 with heart disease and 138 without. Because the two groups are close in size, accuracy is a fair headline metric (it would be misleading if, say, 95% of patients were healthy).

Setting a bar to beat. I started with a baseline that always predicts the most common class. It is right 52% of the time on the test set, so any useful model has to clearly beat 52%.

Comparing models. I held out 30% of patients as a test set the models never saw during training, then compared a decision tree, logistic regression and a random forest. I also ran 10-fold cross-validation, training and testing on ten different splits, so the result does not depend on one lucky split.

The key improvement. Several columns are codes, not quantities: chest pain type 0-3 is a label, so type 3 is not 'three times' type 1. Feeding the codes in as numbers forces the model to treat them as a scale. One-hot encoding turns each code into its own yes/no column, and scaling puts numeric features on the same footing. That lifted logistic regression from 0.81 to 0.85 accuracy.

The result. The final model reaches 0.85 accuracy and 0.91 ROC-AUC, and catches 42 of 47 heart-disease patients in the test set (recall 0.89). Cross-validation gives the same 0.85, so the result is stable.

Why it matters. Logistic regression performed as well as the random forest while staying interpretable: every coefficient shows how a factor moves the risk, which a clinician can check against

medical knowledge.

Explaining predictions with Generative AI. I added a retrieval-augmented generation (RAG) explainer. For each patient it finds the top 3 factors, retrieves the matching passages from a 23-passage knowledge base using sentence-transformer embeddings, and presents a plain-language explanation; a local LLM (Qwen2.5-1.5B) answers follow-up questions in a small chat app.

What evaluation showed. I tested five designs. When the small LLM wrote the explanations itself, it invented numbers and details (for example calling a female patient 'male'). The final design keeps the LLM away from facts: code writes the risk and directions, definitions are quoted from the retrieved passages, and the LLM only answers follow-ups. Retrieval found the right passage 92.7% of the time, and 5 of 5 chat answers contained no invented numbers.

5. Exploratory insights

- Patients are concentrated in middle to late adulthood: average age 54, with 84% aged 40-65.
- About half the patients (151 of 303) have cholesterol above 240 mg/dl, the usual high-risk threshold, and women in this dataset average higher cholesterol than men (261 vs. 239 mg/dl).
- Maximum heart rate falls with age (correlation -0.40) and is higher in the heart-disease group in this dataset (159 vs. 139 bpm).
- Cholesterol barely separates the two groups (correlation with the target -0.09; group averages 242 vs. 251 mg/dl), so it is not a strong predictor here despite its clinical importance. Chest pain type, exercise angina and ST depression separate the groups far better.



Figure: Maximum heart rate by age, coloured by outcome

6. Results

Model	Accuracy	ROC-AUC	Recall	10-fold CV acc.
Majority-class baseline	0.52	0.50	1.00	0.55
Decision Tree	0.75	0.75	0.74	0.75
Logistic Regression (raw)	0.81	0.89	0.89	0.82
Random Forest	0.84	0.91	0.91	0.83
Logistic Regression (scaled + one-hot)	0.85	0.91	0.89	0.85

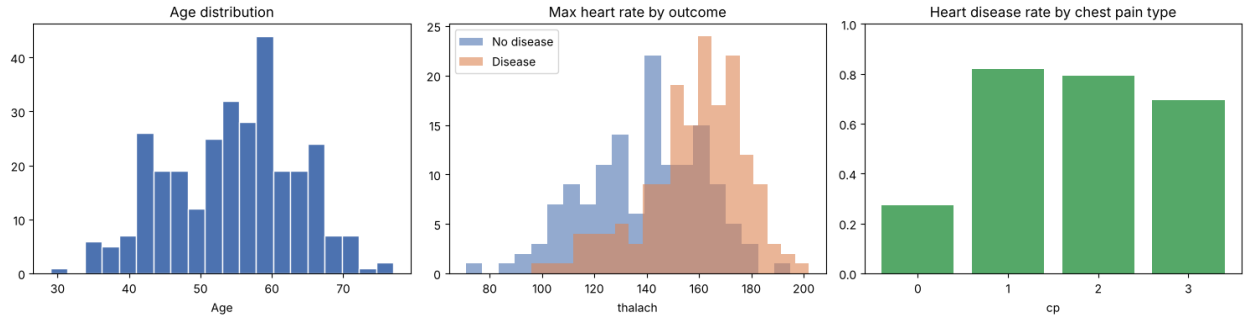


Figure: Age distribution, maximum heart rate by outcome, and disease rate by chest-pain type

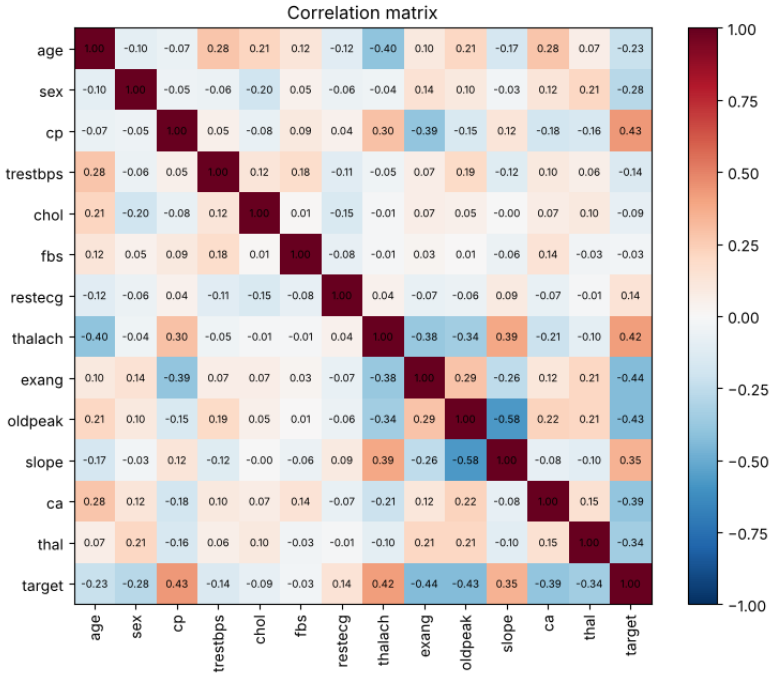


Figure: Correlation matrix of all clinical features and the target

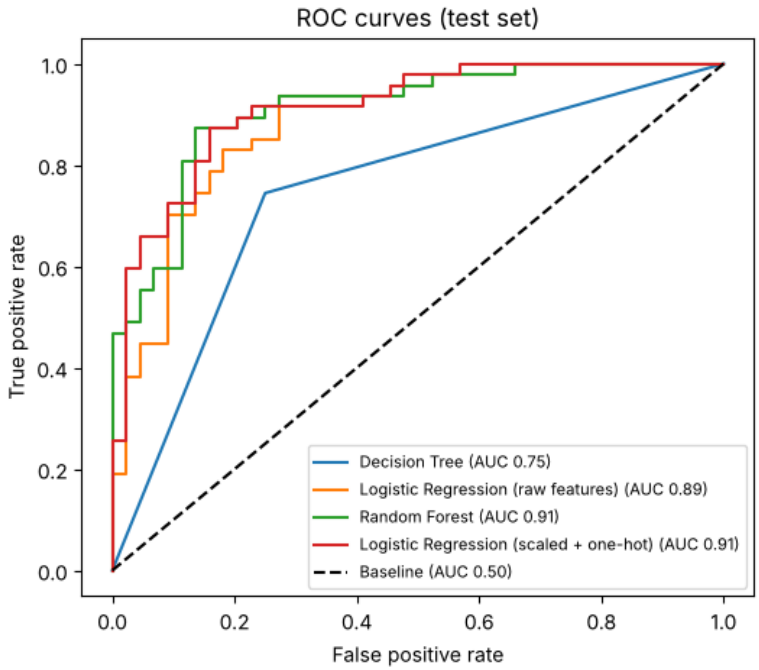


Figure: ROC curves on the test set

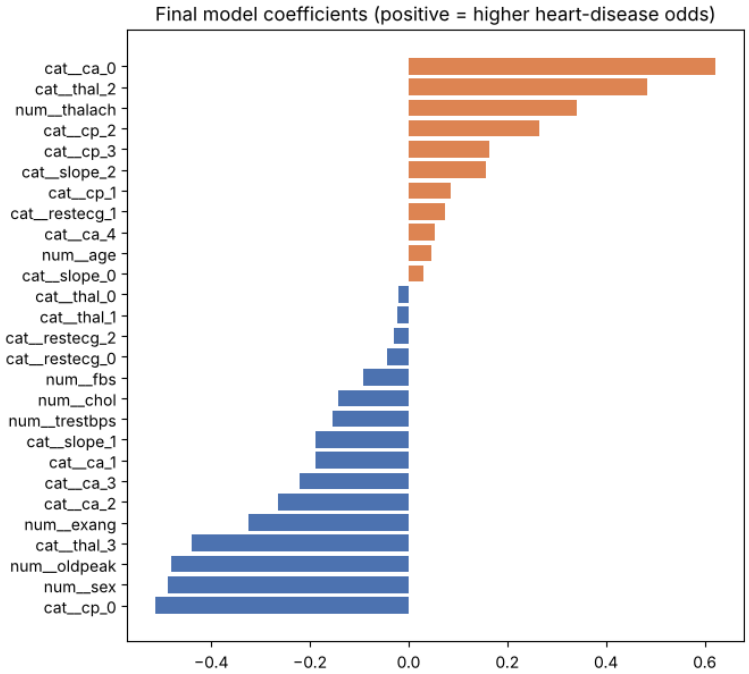


Figure: Final model coefficients: positive values raise the odds of heart disease

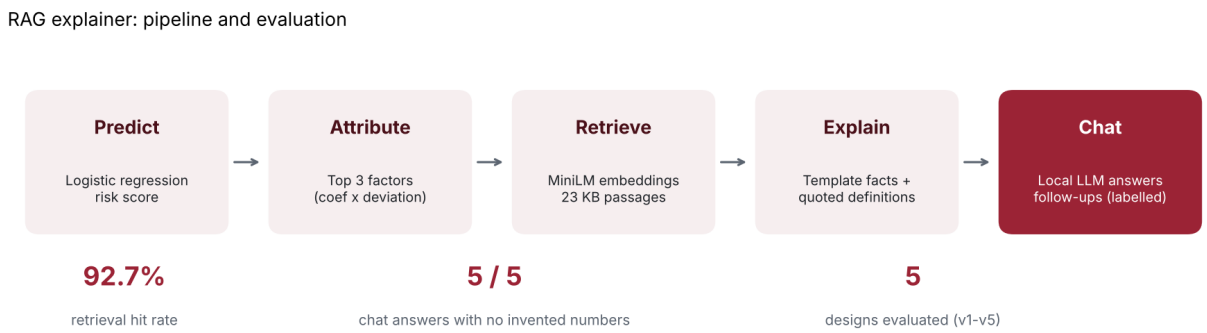


Figure: RAG explainer pipeline and evaluation results

7. Key findings

- Logistic regression improved accuracy from a 0.52 baseline to 0.85, with 0.91 ROC-AUC; 10-fold cross-validation (0.85 accuracy, 0.91 AUC) confirms it is not a lucky split.
- One-hot encoding the categorical clinical codes added about 3 points of accuracy over raw features, and beat the decision tree by 10 points.
- Recall of 0.89: only 5 of 47 heart-disease patients in the test set were missed, which matters most for screening.
- Strongest signals: number of major vessels, chest pain type, thal result, sex, ST depression and maximum heart rate.
- RAG explainer: retrieval found the right knowledge-base passage for 92.7% of patients' top factors. After five design iterations the chat answered 5 of 5 test questions without invented numbers; every earlier free-text version invented statistics or misstated values.

8. How the model could be used

- Early detection: flag at-risk patients from routine measurements so they are referred for further testing sooner.
- Decision support: give clinicians an explainable second opinion, with coefficients showing which factors drove a patient's score.
- Targeted prevention: focus monitoring and lifestyle programmes on the patients ranked highest risk.
- All of these require validation on other hospitals first; this is a screening aid, not a diagnosis.

9. Why these choices

Logistic regression over random forest. Both scored about 0.91 ROC-AUC. When accuracy is equal, the model a clinician can understand wins, and logistic regression coefficients are directly interpretable.

Recall as the metric to watch. In screening, missing a sick patient (false negative) is worse than sending a healthy one for an extra test (false positive). Recall of 0.89 means few true cases are missed.

Cross-validation. With only 303 patients, a single 91-patient test set is noisy. Ten-fold cross-validation checks that the score holds across many splits.

Dropping linear regression. The original course project also tried linear regression, but the target is yes/no, so a classification model is the right tool.

Keeping the LLM away from facts. A 1.5B-parameter model is fluent but unreliable with numbers. Generating facts with code and checking LLM answers for unsupported numbers makes the explainer trustworthy without needing a large paid model.

10. Key terms explained

Baseline	The score of the simplest possible rule (always guess the most common class). It shows how much the model actually adds.
Accuracy	Share of patients classified correctly.
Precision / recall	Precision: of patients flagged as at risk, how many truly have disease. Recall: of patients with disease, how many the model catches.
ROC-AUC	How well the model ranks patients by risk, from 0.5 (random) to 1.0 (perfect). 0.91 means a randomly chosen sick patient gets a higher risk score than a randomly chosen healthy patient 91% of the time.
One-hot encoding	Turning a categorical code into separate yes/no columns so the model does not treat category numbers as quantities.
Coefficient	In logistic regression, how much a feature raises (positive) or lowers (negative) the log-odds of heart disease.
RAG	Retrieval-augmented generation: retrieve relevant documents first, then have the LLM answer using only those documents, which reduces made-up answers.
Hallucination	When an LLM states something not supported by its input, such as an invented statistic.

Tools and skills

Python, pandas, scikit-learn, logistic regression, random forest, cross-validation, ROC-AUC, feature encoding, RAG, sentence-transformers, Hugging Face transformers (Qwen2.5 LLM), Gradio, matplotlib

Code: heart_disease_prediction.ipynb in this folder reproduces every number in this report.